

$$a_n = 3 \longrightarrow a_4 = 3$$

$$a_{n+1} = 7 + a_{n-1} \quad a_{4+1} = a_5 = 7 + a_{4-1}$$

$$7 + a_{5-1}$$

$$7 + a_4$$

$$a_5 = 7 + a_{4-1}$$

$$= 7 + a_4$$

$$7 + 3$$

2.1. Determine el valor de a_4 para la sucesión definida por

$$\begin{cases} a_n = n^2 a_{n-1}, & \text{si } n \geq 2, \\ a_1 = 1. \end{cases}$$

$$a_1 = 1$$

$$a_2 = 2^2 \cdot a_{2-1} = 2^2 \cdot a_1$$

$$4 \cdot 1 = 4 \quad a_2$$

$$a_3 = 3^2 \cdot a_{3-1}$$

$$= 3^2 \cdot a_2$$

$$a_3 = 9 \cdot 4 = \boxed{36}$$

$$\begin{cases} a_n = n^2 a_{n-1}, & \text{si } n \geq 2, \\ a_1 = 1. \end{cases}$$

$$a_4 = 4^2 \cdot a_{4-1}$$

$$= 4^2 \cdot a_3 \rightarrow 4^2 \cdot 36 = \boxed{576}$$

2.3. Determine la fórmula explícita de la sucesión definida por

$$\begin{cases} a_n = 5a_{n-1} + 6a_{n-2}, & \text{si } n \geq 2, \\ a_0 = 2, & a_1 = -1. \end{cases}$$

$$a_n = 5a_{n-1} + 6a_{n-2}$$

$$a_n - 5a_{n-1} - 6a_{n-2} = 0$$

$$x^2 - 5x - 6 = 0$$

$$\begin{array}{rcl} x & \times & -6 = -6x \\ x & \times & 3 = 3x \\ & & -5x \end{array}$$

$$(x-6)(x+1) = 0$$

$$x-6=0 \quad x+1=0$$

$$\begin{array}{cc} x_1 = 6 & x_2 = -1 \\ A & B \end{array}$$

$$a_n = A(x_1)^n + B(x_2)^n \quad a_0 = 2$$

$$a_1 = -1$$

$$a_n = A(6)^n + B(-1)^n \quad a^0 = 1$$

$$a_0 = A \cdot 6^0 + B(-1)^0 = \boxed{A+B=2}$$

$$a_1 = A \cdot 6^1 + B(-1)^1 = \boxed{6A-B=-1}$$

$$\begin{cases} A+B = 2 \\ 6A-B = -1 \end{cases}$$

$$A = \frac{1}{7} \quad B = \frac{13}{7} \quad \begin{array}{cc} a_n = A(x_1)^n + B(x_2)^n \\ x_1 = 6 & x_2 = -1 \end{array}$$

$$\boxed{a_n = \frac{1}{7} 6^n + \frac{13}{7} (-1)^n}$$

2.5. Determine la fórmula explícita de la sucesión definida por

$$\begin{cases} a_n = -6a_{n-1} - 9a_{n-2}, & \text{si } n \geq 2, \\ a_0 = 4, & a_1 = -15. \end{cases}$$

$$a_n + 6a_{n-1} + 9a_{n-2} = 0$$

$$x^2 + 6x + 9 \rightarrow x = \frac{-2}{3}$$

$$\begin{array}{rcl} x & \times & 3 = 3x \\ x & \times & 3 = 3x \\ & & \hline & & 6x \end{array}$$

$$(x+3)(x+3)$$

$$x_1 = -3 \quad x_2 = -3 \quad \swarrow \text{Por que son repetidos}$$

$$a_n = A(x_1)^n + B \cdot n \cdot (x_2)^n$$

$$a_0 = A(-3)^0 + B \cdot 0 \cdot (-3)^0$$

$$A = 4$$

$$a_0 = 4$$

$$a_1 = -15$$

$$a_1 = A(-3)^1 + B \cdot 1 \cdot (-3)^1$$

$$-3A - 3B = -15$$

$$A + 0B = 4$$

$$-3A - 3B = -15$$

$$A = 4 \quad B = 1$$

$$a_n = A(x_1)^n + B \cdot n \cdot (x_2)^n \quad A = 4 \quad B = 1$$

$$x_1 = -3 \quad x_2 = -3$$

$$\boxed{a_n = 4 \cdot (-3)^n + n(-3)^n}$$

2.10. Determine la fórmula por recurrencia para la sucesión dada en forma explícita por

$$U_n = 2(-3)^n - 5n(-3)^n, \quad n \geq 1.$$

$$a_n = A \cdot (x_1)^n + B \cdot n (x_2)^n$$

$$x_1 = -3 \quad x_2 = -3$$

$$x + 3 = 0 \quad x + 3 = 0$$

$$(x + 3)(x + 3) = 0$$

$$(x + 3)^2 \rightarrow a^2 + 2ab + b^2$$

$$x^2 + 2 \cdot x \cdot 3 + 3^2$$

$$x^2 + 6x + 9$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$a_n \quad a_{n-1} \quad a_{n-2}$$

$$a_n + 6a_{n-1} + 9a_{n-2} = 0$$

$$a_n = -6a_{n-1} - 9a_{n-2} \quad a_n = 2(-3)^n - 5n(-3)^n$$

$$a_1 = 2(-3)^1 - 5 \cdot 1(-3)^1 = 9$$

$$a_2 = 2(-3)^2 - 5 \cdot 2(-3)^2 = -72$$

$$\boxed{\begin{array}{l} a_n = -6a_{n-1} - 9a_{n-2} \\ a_1 = 9 \quad a_2 = -72 \end{array}}$$

2.11. Determine la fórmula por recurrencia para la sucesión dada en forma explícita por

$$a_n = 5 + (-2)^n, \quad n \geq 0.$$

$$a_n = A(x_1)^n + B(x_2)^n \quad x_1 \neq x_2$$

$$\triangleright a_n = 5(1)^n + (-2)^n \quad 1^n = 1$$

$$x_1 = 1 \quad x_2 = -2$$

$$x - 1 = 0 \quad x + 2 = 0$$

$$(x - 1)(x + 2) = 0$$

$$x^2 + 2x - x - 2 = 0$$

$$x^2 + x - 2 = 0$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$a_n \quad a_{n-1} \quad a_{n-2}$$

$$a_n + a_{n-1} - 2a_{n-2} = 0$$

$$a_n = -a_{n-1} + 2a_{n-2}$$

$$a_n = 5(1)^n + (-2)^n \quad a_0, a_1$$

$$a_0 = 5(1)^0 + (-2)^0 = 6$$

$$a_1 = 5(1)^1 + (-2)^1 = 3$$

$$\boxed{a_n = -a_{n-1} + 2a_{n-2} \\ a_0 = 6 \quad a_1 = 3}$$

1.1. Use el método de inducción matemática para demostrar que

$$1 + 2 + 3 + \dots + n \leq \frac{(2n+1)^2}{8}, \quad \forall n \in \mathbb{N}, n \geq 1.$$

$$\begin{matrix} n \\ \leq \\ k=1 \end{matrix} \quad k \leq \frac{(2n+1)^2}{8}$$

$$n=1 \quad 1 \leq \frac{(2 \cdot 1 + 1)^2}{8}$$

$$1 \leq \frac{9}{8} \quad \checkmark$$

$$\begin{matrix} n=p \\ \leq \\ k=1 \end{matrix} \quad k \leq \frac{(2p+1)^2}{8} \text{ H.I.} \quad \begin{matrix} 2(p+1)+1 \\ 2p+2+1 \\ 2p+3 \end{matrix}$$

$$\begin{matrix} n=p+1 \\ \leq \\ k=1 \end{matrix} \quad k \leq \frac{(2p+3)^2}{8} \text{ H.Q.D.}$$

$$\frac{(2p+3)^2}{8} \rightarrow \frac{4p^2 + 12p + 9}{8} \quad \swarrow a^2 + 2ab + b^2$$

$$\begin{matrix} p+1 \\ \leq \\ k=1 \end{matrix} \quad k \Rightarrow \begin{matrix} p \\ \leq \\ k=1 \end{matrix} \quad k + (p+1)$$

$$\frac{(2p+1)^2}{8} + (p+1) \text{ H.I.}$$

$$\frac{(2p+1)^2}{8} + (p+1) \cdot \frac{8}{8}$$

$$\frac{(2p+1)^2 + 8p + 8}{8}$$

$$\frac{4p^2 + 4p + 1 + 8p + 8}{8}$$

$$\frac{4p^2 + 12p + 9}{8}$$

$$\boxed{\frac{(2p+3)^2}{8}}$$